

SOFIA LOYA

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EDUCATION

Carnegie Mellon University

Master of Science Mechanical Engineering | GPA: 4.0/4.0

Rales Fellows Program

Selected Coursework: [Machine Learning and Artificial Intelligence for Engineers, Introduction to Deep Learning]

Texas A&M University

Bachelor of Science in Mechanical Engineering | GPA: 3.587/4.0

Pittsburgh, PA

May 2027

College Station, TX

May 2021

PROFESSIONAL EXPERIENCE

Medtronic

Quality Engineering Intern | Cardiac Ablation Solutions

Mounds View, MN

June 2026 – Aug 2026

- Drove the assessment and development of seven test methods and characterization tests for a small focal catheter to comply with specifications for EU Clinical Study
- Automated measurement collection on LCR Meter for capacitive coupling test method that reduced 99 manual data entry points to one data file

Dell Technologies

Mechanical Engineer II | Dell Commercial Mobile Latitude

Austin, TX

April 2023 – June 2025

- Investigated hinge field failures through adhesive performance evaluation, mechanical stress analysis, and fatigue testing to develop internal specifications mitigating a \$500M USD reliability risk
- Conducted ANSYS simulations comparing injection-molded hinge designs to identify geometry improvements that reduce stress concentrations and guided future design standards
- Created weight reduction opportunities via mechanical redesign in CAD and material selection, decreasing weight of single platform by 10% while maintaining stiffness and reliability targets
- Reviewed 8 bills of materials with 200+ parts each and drove design changes to reduce cost by 12M USD
- Diagnosed eDP cable design inconsistencies across platforms and partnered with design, electrical, and manufacturing teams to simplify configurations, achieving a 15% reduction in part complexity
- Initiated and developed 'Recent Graduate CREO Learning' by detailing a CAD development path for five new hires

Mechanical Engineer I | Dell Commercial Mobile Rugged

June 2021 – April 2023

- Developed and filed patent for electromechanical intrusion-prevention design integrating precision sensor and secure mechanical engagement
- Designed I/O port housing in CREO and analyzed in ANSYS, reducing testing time by one week
- Built a Power BI comparison dashboard of mobile platforms and competitors, streamlining executive access to competitor insights and supporting faster strategic decision-making
- Led cross-functional team for design competition and pitched a framework solution aimed at turning a \$240 million security business into \$2 billion business

Mechanical Engineer Intern | Dell Desktop Precision

May 2020 – August 2020

- Prototyped housing for removable SSD cards in Creo for Precision tower quadrupling memory capacity
- Collaborated with engineering team to design module that increases capacity of Precision towers by 8 TB

Ethicon | Laparoscopic Surgical Products

R&D Intern

Cincinnati, OH

January 2019 – May 2019

- Designed and built a test fixture for laparoscopic surgery simulation, applying mechanics and thermodynamics to maintain 1 psi pressure, enabling realistic use of laparoscopic devices
- Led verification testing for 40 medical devices, applying mechanical reliability and statistical analysis to validate compliance with shelf-life and performance standards

RESEARCH EXPERIENCE

Biohybrid and Organic Robotics Group – Carnegie Mellon University

- Conducting research on compliant robotic actuators and soft material integration for bioinspired robotic systems

SKILLS

SolidWorks, Creo, Python, Matlab, Finite Element Analysis, LabVIEW, applied statistics, Cyberbotics: Webot, PowerBI, reliability testing, rapid prototyping (FDM technologies), JIRA, Agile, and data analytics